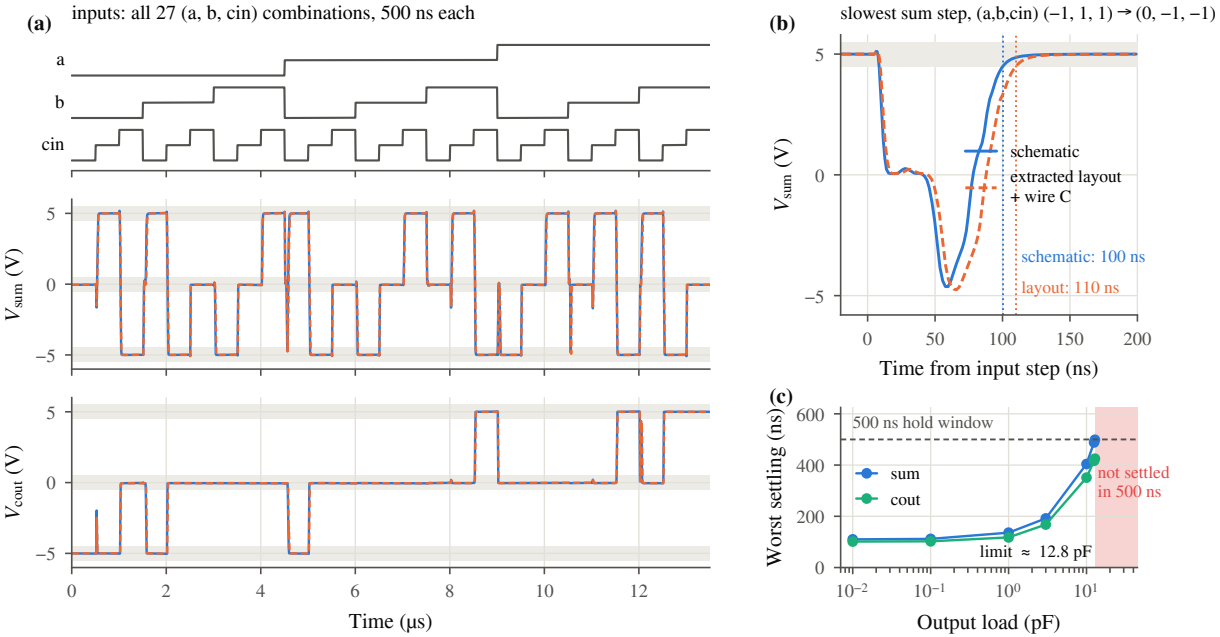
(a)
inputs: all 27 (a, b, cin) combinations, 500 ns each
a
b
cin
V_{sum} (V)
V_{cout} (V)
5
0
−5
0
2
4
6
8
10
12
Time (µs)
(b)
slowest sum step, (a,b,cin) (−1, 1, 1) → (0, −1, −1)
schematic
extracted layout
+ wire C
schematic: 100 ns
layout: 110 ns
0
50
100
150
200
Time from input step (ns)
(c)
500 ns hold window
sum
cout
not settled
in 500 ns
limit ≈ 12.8 pF
Worst settling (ns)
600
400
200
10^{-2}
10^{-1}
10^{0}
10^{1}
Output load (pF)